

David Franz

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Summary

Full-stack software engineer focused on AI and LLM applications in Python, JVM backends, and TypeScript frontends, with interests in compilers, algorithms, and formal methods. I enjoy building reliable APIs, visual tooling, and language infrastructure. Outside of software, I enjoy hiking, making music, and game development.

Skills

- **Languages:** Java, Kotlin, JavaScript, TypeScript, Python, C, C++
- **Frontend:** React, Angular, Knockout, HTML, CSS
- **Backend:** Spring, Spring Boot, Node, Express, Vert.x, FastAPI
- **ML/AI:** PyTorch, TensorFlow, Keras, scikit-learn, LoRA, HuggingFace, LangChain, NLTK
- **DevOps/Platforms:** Linux, Git, Docker, Terraform, AWS, Azure
- **Databases:** PostgreSQL, Alembic, MongoDB, MariaDB

Experience

Technical Consultant — Palo IT

Jan 2026 — Present · Sydney

Stack: Python, PostgreSQL, Alembic, AWS (Lambda, SQS), Terraform, React, TypeScript
Software engineering consultant working with a major Australian radiology company to develop a new RIS (Radiology Information System).

- Contributed to the system design of the RIS, helping design a variety of key services and features end to end.
- Worked with key stakeholders to gather requirements and turn them into solutions.
- Built and maintained the core data service in Python and PostgreSQL (Alembic), used for authoring data consumed by other services and kept in sync via SQS and a transactional outbox; deployed as a Lambda on AWS.
- Built and maintained a room note classifier service that dynamically ingests notes from the existing RIS calendar and converts raw notes into rule assignments, staff assignments, and room blocks.
- Built a responsive React application for managing the core data service.
- Built a data seed ingestion pipeline from static CSV and API data.

- Built Terraform pipelines for deploying services.

Software Engineer — Servicely

July 2024 – April 2025 · Sydney

Stack: Java, Kotlin, Spring, PostgreSQL, TypeScript, Angular, Knockout, AWS

- Maintained and enhanced existing backend REST APIs.
- Developed and maintained mobile REST APIs in Java/Kotlin with Spring.
- Migrated Java APIs to Kotlin to improve maintainability and readability.
- Implemented new Angular pages and components with cross-device responsiveness.
- Migrated legacy Knockout/HTML/JS pages and renderers to modern Angular.

Product Software Engineer — Solnet Solutions

Nov 2021 – May 2024 · Wellington

Stack: Java, Vert.x, MariaDB, TypeScript, React, Docker, Azure

- Built a formal specification language with expression evaluation, type definitions, variable/object handling, subprocess management, grammar implementation, and AST generation/manipulation.
- Implemented a grammar-based natural-language system to create propositional logic.
- Integrated backend version-control features for a web IDE using JGit and Vert.x.
- Delivered complex React components, including a visualization tool for version-control histories.
- Managed WebSocket APIs for real-time client-server communication.

Research Assistant — Victoria University

Nov 2020 – Feb 2021 · Wellington

Stack: Racket, Redex

- Translated concepts from an academic paper into a working prototype.
- Implemented formal syntax and semantics of a research-defined language in Racket/Redex leveraging temporal logic for program verification.
- Gained hands-on experience with formal methods, verification, and secure system design.

Projects

- **ctx-sys** — Local hybrid RAG over a code knowledge graph. Indexes a codebase with tree-sitter, embeddings, and a relationship graph, then serves precise context to any AI assistant over MCP — fusing keyword (FTS5), vector, and graph search with reciprocal rank fusion. Local-first via Ollama.
Stack: TypeScript, Node, SQLite (FTS5, sqlite-vec), tree-sitter, Ollama, MCP.

- **yaao** — Plans, converts, and runs multi-agent software work across parallel git worktrees, with dependency-aware branching, topological merge-back, and validation-gated merges. Agent-agnostic (Claude Code, Cursor, Copilot, Codex, raw API) and MCP-first; ships a CLI, an MCP server, and a live web viewer.
Stack: TypeScript, Node, Hono, MCP, Git.
- **flowlang** — Compiler for a small, strongly-typed JVM language (ANTLR4 grammar → typed AST → bytecode via ASM) blending functional collections, structural records, pattern matching, and declarative task orchestration. Includes a web playground backed by a Spring Boot run API, with hundreds of categorized tests.
Stack: Java, ANTLR, ASM, Spring Boot, React, TypeScript.
- **strobe** — Live-coding music library: patterns and effects are written in TypeScript using the TidalCycles pattern model and hot-reloaded while playing, while a Rust engine renders synthesis, a per-track effects rack, and sampling on a real-time audio thread.
Stack: Rust, TypeScript.
- **Meridian** — Deep-simulation city builder inspired by Cities: Skylines and SimCity. Simulates cities down to the household and models each trip — mode choice, transfers, congestion — across a region of cities that trade and commute. Soundtrack generated programmatically with strobe.
Stack: C#, Godot, Blender, Python.
- **CNN Skin Cancer Detector** — CNNs for classifying skin lesions in the ISIC dataset into 9 conditions, grouped as cancerous, precancerous, or benign. Reproduced a baseline CNN from the literature, then designed a parallel-kernel CNN (five kernel sizes, 3 to 27) that raised 9-class validation accuracy from 58.9% to 63.2% and cancer recall from 79% to 84%. Accompanied by a written report covering false negatives, dataset bias, and the safety risks of the approach.
Stack: Python, PyTorch, Jupyter, Google Colab.

Education

- **BSc in Computer Science & Mathematics** — Victoria University of Wellington, 2021